Compatibilidad: Occupancy API ↔ InnoBus

Comparación de los records de InnoBus (`backend/InnoBus.Api/Ocupacion/ContratosOcupacion.cs`) con respuestas reales de la Occupancy API publicada (https://lg0618pp-002-site2.htempurl.com), tomadas el 07/10/2026 a las 12:41 (hora local).

Cada fila es una propiedad de un record de InnoBus: el campo JSON que ASP.NET Core le asigna (camelCase), si viene en la respuesta, su tipo y un valor de ejemplo. «Sí» en la última columna significa que el valor se deserializa en el tipo C# declarado.

Resultado: 118 de 118 propiedades coinciden: no falta ningún campo y ningún tipo es distinto. Se revisaron con datos reales los 18 records de InnoBus.

`GET /api/v1/vehicles`

Record	Propiedad C#	Tipo C#	Campo JSON	Tipo JSON	Ejemplo	Coincide
VehicleList	`Vehicles`	`IReadOnlyList<VehicleState>`	`vehicles`	array	`[43 elementos]`	**Sí**
VehicleState	`VehicleId`	`string`	`vehicleId`	string	`"SJB-10356"`	**Sí**
VehicleState	`AsOf`	`DateTimeOffset`	`asOf`	string	`"2026-10-07T18:41:37.9717188+00:00"`	**Sí**
VehicleState	`Stale`	`bool`	`stale`	boolean	`false`	**Sí**
VehicleState	`Location`	`GeoLocation?`	`location`	object	{…}	**Sí**
GeoLocation	`Lat`	`double`	`lat`	number (decimal)	`9.948412`	**Sí**
GeoLocation	`Lon`	`double`	`lon`	number (decimal)	`-84.050266`	**Sí**
GeoLocation	`SpeedKmh`	`double?`	`speedKmh`	number (entero)	`23`	**Sí**
GeoLocation	`HeadingDeg`	`double?`	`headingDeg`	number (decimal)	`61.2`	**Sí**
VehicleState	`Trip`	`TripInfo?`	`trip`	object	{…}	**Sí**
TripInfo	`RouteId`	`string?`	`routeId`	string	`"R142"`	**Sí**
TripInfo	`Direction`	`string?`	`direction`	string	`"OUTBOUND"`	**Sí**
VehicleState	`Occupancy`	`OccupancySnapshot?`	`occupancy`	object	{…}	**Sí**
OccupancySnapshot	`PassengerCount`	`int`	`passengerCount`	number (entero)	`5`	**Sí**
OccupancySnapshot	`Capacity`	`int`	`capacity`	number (entero)	`90`	**Sí**
OccupancySnapshot	`Percent`	`int`	`percent`	number (entero)	`6`	**Sí**
OccupancySnapshot	`Status`	`string`	`status`	string	`"MANY_SEATS_AVAILABLE"`	**Sí**
OccupancySnapshot	`Source`	`string`	`source`	string	`"SIMULATED"`	**Sí**

Record	Propiedad C#	Tipo C#	Campo JSON	Tipo JSON	Ejemplo	Coincide
OccupancySnapshot	`MeasuredAt`	`DateTimeOffset`	`measuredAt`	string	`"2026-10-07T18:41:37.9717188+00:00"`	**Sí**
VehicleState	`Device`	`DeviceHealth?`	`device`	object	{…}	**Sí**
DeviceHealth	`Online`	`bool`	`online`	boolean	`true`	**Sí**
DeviceHealth	`LastSeen`	`DateTimeOffset`	`lastSeen`	string	`"2026-10-07T18:41:37.9717479+00:00"`	**Sí**
DeviceHealth	`CameraOnline`	`bool?`	`cameraOnline`	boolean	`true`	**Sí**
DeviceHealth	`Firmware`	`string?`	`firmware`	string	`"sim-2.0"`	**Sí**

`GET /api/v1/vehicles/{placa}`

Record	Propiedad C#	Tipo C#	Campo JSON	Tipo JSON	Ejemplo	Coincide
VehicleState	`VehicleId`	`string`	`vehicleId`	string	`"SJB-10356"`	**Sí**
VehicleState	`AsOf`	`DateTimeOffset`	`asOf`	string	`"2026-10-07T18:41:42.9717286+00:00"`	**Sí**
VehicleState	`Stale`	`bool`	`stale`	boolean	`false`	**Sí**
VehicleState	`Location`	`GeoLocation?`	`location`	object	{…}	**Sí**
GeoLocation	`Lat`	`double`	`lat`	number (decimal)	`9.948558`	**Sí**
GeoLocation	`Lon`	`double`	`lon`	number (decimal)	`-84.05`	**Sí**
GeoLocation	`SpeedKmh`	`double?`	`speedKmh`	number (entero)	`24`	**Sí**
GeoLocation	`HeadingDeg`	`double?`	`headingDeg`	number (decimal)	`61.2`	**Sí**
VehicleState	`Trip`	`TripInfo?`	`trip`	object	{…}	**Sí**
TripInfo	`RouteId`	`string?`	`routeId`	string	`"R142"`	**Sí**
TripInfo	`Direction`	`string?`	`direction`	string	`"OUTBOUND"`	**Sí**
VehicleState	`Occupancy`	`OccupancySnapshot?`	`occupancy`	object	{…}	**Sí**
OccupancySnapshot	`PassengerCount`	`int`	`passengerCount`	number (entero)	`5`	**Sí**
OccupancySnapshot	`Capacity`	`int`	`capacity`	number (entero)	`90`	**Sí**
OccupancySnapshot	`Percent`	`int`	`percent`	number (entero)	`6`	**Sí**
OccupancySnapshot	`Status`	`string`	`status`	string	`"MANY_SEATS_AVAILABLE"`	**Sí**
OccupancySnapshot	`Source`	`string`	`source`	string	`"SIMULATED"`	**Sí**

Record	Propiedad C#	Tipo C#	Campo JSON	Tipo JSON	Ejemplo	Coincide
OccupancySnapshot	`MeasuredAt`	`DateTimeOffset`	`measuredAt`	string	`"2026-10-07T18:41:42.9717286+00:00"`	**Sí**
VehicleState	`Device`	`DeviceHealth?`	`device`	object	{…}	**Sí**
DeviceHealth	`Online`	`bool`	`online`	boolean	`true`	**Sí**
DeviceHealth	`LastSeen`	`DateTimeOffset`	`lastSeen`	string	`"2026-10-07T18:41:42.9717538+00:00"`	**Sí**
DeviceHealth	`CameraOnline`	`bool?`	`cameraOnline`	boolean	`true`	**Sí**
DeviceHealth	`Firmware`	`string?`	`firmware`	string	`"sim-2.0"`	**Sí**

`GET /api/v1/vehicles/{placa}/history?interval=5m`

Record	Propiedad C#	Tipo C#	Campo JSON	Tipo JSON	Ejemplo	Coincide
OccupancyHistory	`VehicleId`	`string`	`vehicleId`	string	`"SJB-10356"`	**Sí**
OccupancyHistory	`From`	`DateTimeOffset`	`from`	string	`"2026-10-07T17:41:43.9547001+00:00"`	**Sí**
OccupancyHistory	`To`	`DateTimeOffset`	`to`	string	`"2026-10-07T18:41:43.9547001+00:00"`	**Sí**
OccupancyHistory	`Interval`	`string`	`interval`	string	`"5m"`	**Sí**
OccupancyHistory	`Points`	`IReadOnlyList<HistoryPoint>`	`points`	array	`[12 elementos]`	**Sí**
HistoryPoint	`Time`	`DateTimeOffset`	`time`	string	`"2026-10-07T17:41:43.9547001+00:00"`	**Sí**
HistoryPoint	`PassengerCount`	`int`	`passengerCount`	number (entero)	`14`	**Sí**
HistoryPoint	`PeakPassengerCount`	`int`	`peakPassengerCount`	number (entero)	`14`	**Sí**
HistoryPoint	`Capacity`	`int`	`capacity`	number (entero)	`90`	**Sí**
HistoryPoint	`Percent`	`int`	`percent`	number (entero)	`16`	**Sí**

`GET /api/v1/vehicles/{placa}/events?since=`

Record	Propiedad C#	Tipo C#	Campo JSON	Tipo JSON	Ejemplo	Coincide
VehicleEventList	`VehicleId`	`string`	`vehicleId`	string	`"SJB-10356"`	**Sí**
VehicleEventList	`Events`	`IReadOnlyList<DoorEvent>`	`events`	array	`[16 elementos]`	**Sí**

Record	Propiedad C#	Tipo C#	Campo JSON	Tipo JSON	Ejemplo	Coincide
DoorEvent	`Door`	`int`	`door`	number (entero)	`1`	**Sí**
DoorEvent	`Boardings`	`int`	`boardings`	number (entero)	`1`	**Sí**
DoorEvent	`Alightings`	`int`	`alightings`	number (entero)	`0`	**Sí**
DoorEvent	`Location`	`GeoLocation?`	`location`	object	{…}	**Sí**
GeoLocation	`Lat`	`double`	`lat`	number (decimal)	`9.937322`	**Sí**
GeoLocation	`Lon`	`double`	`lon`	number (decimal)	`-84.070059`	**Sí**
GeoLocation	`SpeedKmh`	`double?`	`speedKmh`	number (entero)	`0`	**Sí**
GeoLocation	`HeadingDeg`	`double?`	`headingDeg`	null	`null`	**Sí**
DoorEvent	`OpenedAt`	`DateTimeOffset`	`openedAt`	string	`"2026-10-07T18:11:32.9639116+00:00"`	**Sí**
DoorEvent	`ClosedAt`	`DateTimeOffset`	`closedAt`	string	`"2026-10-07T18:11:52.9639116+00:00"`	**Sí**
DoorEvent	`OccupancyAfter`	`int?`	`occupancyAfter`	number (entero)	`15`	**Sí**

`GET /api/v1/trips/plan`

Record	Propiedad C#	Tipo C#	Campo JSON	Tipo JSON	Ejemplo	Coincide
TripPlan	`From`	`GeoPoint`	`from`	object	{…}	**Sí**
GeoPoint	`Lat`	`double`	`lat`	number (decimal)	`9.933`	**Sí**
GeoPoint	`Lon`	`double`	`lon`	number (decimal)	`-84.079`	**Sí**
TripPlan	`To`	`GeoPoint`	`to`	object	{…}	**Sí**
TripPlan	`GeneratedAt`	`DateTimeOffset`	`generatedAt`	string	`"2026-10-07T18:41:43.4391331+00:00"`	**Sí**
TripPlan	`Options`	`IReadOnlyList<TripOption>`	`options`	array	`[1 elementos]`	**Sí**
TripOption	`DepartAt`	`DateTimeOffset`	`departAt`	string	`"2026-10-07T18:41:43.4391331+00:00"`	**Sí**
TripOption	`ArriveAt`	`DateTimeOffset`	`arriveAt`	string	`"2026-10-07T19:04:32.237539+00:00"`	**Sí**
TripOption	`DurationMinutes`	`double`	`durationMinutes`	number (decimal)	`22.8`	**Sí**
TripOption	`Transfers`	`int`	`transfers`	number (entero)	`0`	**Sí**
TripOption	`Legs`	`IReadOnlyList<TripLeg>`	`legs`	array	`[3 elementos]`	**Sí**

Record	Propiedad C#	Tipo C#	Campo JSON	Tipo JSON	Ejemplo	Coincide
TripLeg	`Type`	`string`	`type`	string	`"BUS"`	**Sí**
TripLeg	`From`	`TripPlace`	`from`	object	{…}	**Sí**
TripPlace	`Name`	`string`	`name`	string	`"San José"`	**Sí**
TripPlace	`Lat`	`double`	`lat`	number (decimal)	`9.932732`	**Sí**
TripPlace	`Lon`	`double`	`lon`	number (decimal)	`-84.079187`	**Sí**
TripPlace	`StopId`	`string?`	`stopId`	string	`"R142-00"`	**Sí**
TripLeg	`To`	`TripPlace`	`to`	object	{…}	**Sí**
TripLeg	`DepartAt`	`DateTimeOffset`	`departAt`	string	`"2026-10-07T18:44:13.4391331+00:00"`	**Sí**
TripLeg	`ArriveAt`	`DateTimeOffset`	`arriveAt`	string	`"2026-10-07T18:58:07.0042799+00:00"`	**Sí**
TripLeg	`Minutes`	`double`	`minutes`	number (decimal)	`13.9`	**Sí**
TripLeg	`Meters`	`int`	`meters`	number (entero)	`3631`	**Sí**
TripLeg	`Bus`	`BusRide?`	`bus`	object	{…}	**Sí**
BusRide	`RouteId`	`string`	`routeId`	string	`"R142"`	**Sí**
BusRide	`RouteName`	`string`	`routeName`	string	`"Ruta 142"`	**Sí**
BusRide	`Direction`	`string`	`direction`	string	`"OUTBOUND"`	**Sí**
BusRide	`Headsign`	`string`	`headsign`	string	`"Terminal de Coronado"`	**Sí**
BusRide	`VehicleId`	`string`	`vehicleId`	string	`"SJB-12315"`	**Sí**
BusRide	`WaitMinutes`	`double`	`waitMinutes`	number (decimal)	`1.9`	**Sí**
BusRide	`Stops`	`int`	`stops`	number (entero)	`9`	**Sí**
BusRide	`Vehicle`	`BusNow`	`vehicle`	object	{…}	**Sí**
BusNow	`Location`	`GeoPoint`	`location`	object	{…}	**Sí**
BusNow	`SpeedKmh`	`double?`	`speedKmh`	number (entero)	`0`	**Sí**
BusNow	`HeadingDeg`	`double?`	`headingDeg`	null	`null`	**Sí**
BusNow	`PassengerCount`	`int?`	`passengerCount`	number (entero)	`13`	**Sí**
BusNow	`Capacity`	`int?`	`capacity`	number (entero)	`90`	**Sí**
BusNow	`Percent`	`int?`	`percent`	number (entero)	`14`	**Sí**
BusNow	`Status`	`string?`	`status`	string	`"MANY_SEATS_AVAILABLE"`	**Sí**

Record	Propiedad C#	Tipo C#	Campo JSON	Tipo JSON	Ejemplo	Coincide
BusNow	`DistanceToStopMeters`	`int`	`distanceToStopMeters`	number (entero)	`0`	**Sí**
BusRide	`LaterBuses`	`IReadOnlyList<UpcomingBus>`	`laterBuses`	array	`[2 elementos]`	**Sí**
UpcomingBus	`VehicleId`	`string`	`vehicleId`	string	`"SJB-9581"`	**Sí**
UpcomingBus	`ArriveAt`	`DateTimeOffset`	`arriveAt`	string	`"2026-10-07T18:46:43.4391331+00:00"`	**Sí**
UpcomingBus	`PassengerCount`	`int?`	`passengerCount`	number (entero)	`14`	**Sí**
UpcomingBus	`Capacity`	`int?`	`capacity`	number (entero)	`90`	**Sí**
UpcomingBus	`Status`	`string?`	`status`	string	`"MANY_SEATS_AVAILABLE"`	**Sí**
BusRide	`Path`	`IReadOnlyList<GeoPoint>`	`path`	array	`[13 elementos]`	**Sí**
TripOption	`Warnings`	`IReadOnlyList<string>`	`warnings`	array	`[2 elementos]`	**Sí**
TripPlan	`Message`	`string?`	`message`	null	`null`	**Sí**